1. Amazon CloudWatch — 30 Important Interview Questions & Answers

1. What is Amazon CloudWatch?
Answer:
Amazon CloudWatch is an AWS monitoring and observability service. It helps us monitor AWS resources, applications, logs, metrics, and events. We can use CloudWatch to understand what is happening in our AWS environment and create alerts when something goes wrong.
For example:
If an EC2 server CPU becomes very high, CloudWatch can monitor the CPU and send an alert when it crosses a defined limit.
Example:
EC2
↓
CloudWatch
↓
CPU Metric
↓
Alarm
↓
Notification
Simple ga gurthupettuko:
CloudWatch = Monitor + Logs + Metrics + Alarms

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2. Why do we use CloudWatch?
Answer:
We use CloudWatch to monitor AWS resources and applications. It helps us find performance problems, create alarms, collect logs, and automate actions based on certain conditions.
For example:
If an EC2 instance reaches 80% CPU for a long time, CloudWatch can send a notification to the team.

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3. What are CloudWatch Metrics?
Answer:
Metrics are numerical measurements collected over time. They help us understand the health and performance of AWS resources and applications.
Common metrics include:
CPU utilization
Network traffic
Disk-related metrics
Database connections
Request count
For example:
EC2
↓
CPU Utilization
↓
CloudWatch Metric

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4. What is a CloudWatch Alarm?
Answer:
A CloudWatch Alarm watches a metric and checks whether it crosses a configured condition. If the condition is met, the alarm changes state and can trigger an action such as sending a notification.
For example:
CPU > 80%
↓
CloudWatch Alarm
↓
SNS Notification

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5. What are CloudWatch Logs?
Answer:
CloudWatch Logs is used to collect and store log messages from applications, AWS services, and other supported sources. We can search and analyze these logs to troubleshoot problems.
For example:
If an application gives an error, we can send application logs to CloudWatch Logs and check what caused the problem.

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6. What is a Log Group?
Answer:
A Log Group is a container that holds related CloudWatch log streams. We normally create a log group for a particular application, service, or environment.
For example:
Log Group ↓ Application Logs ↓Log Streams

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7. What is a Log Stream?
Answer:
A Log Stream is a sequence of log events from a specific source. Multiple log streams can exist inside one log group.
For example:
Log Group
│
├── EC2-1 Logs
├── EC2-2 Logs
└── EC2-3 Logs

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8. What is a CloudWatch Dashboard?
Answer:
A CloudWatch Dashboard gives us a visual view of important metrics and information. We can add graphs, numbers, and other monitoring information to one dashboard.
For example:
CloudWatch Dashboard
│
├── EC2 CPU
├── Network
├── RDS Connections
└── Application Errors

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9. What is CloudWatch Event?
Answer:
CloudWatch Events was the earlier name for the event functionality that is now mainly provided through **Amazon EventBridge**. It is used to react when something happens in AWS or an application.
For example:
When an EC2 instance changes state, an event can trigger another action.

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10. What is EventBridge in relation to CloudWatch?
Answer:
Amazon EventBridge is the AWS service used for event-driven architectures and is the successor to CloudWatch Events. It receives events and can route them to targets based on rules.
For example:
AWS Event
↓
EventBridge Rule
↓
Lambda

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11. What is CloudWatch Agent?
Answer:
The CloudWatch Agent is software that can be installed on EC2 or supported servers to collect additional metrics and logs. It can collect information that basic monitoring may not provide by default.
For example:
We can use the CloudWatch Agent to collect memory usage from an EC2 instance.
Example:
EC2
↓
CloudWatch Agent
↓
Memory / Disk / Logs
↓
CloudWatch

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12. What is Basic Monitoring?
Answer:
Basic monitoring provides standard AWS resource metrics at a standard monitoring frequency. It is available for many AWS services without needing an additional agent.
For example:
EC2 can provide basic CPU and network-related metrics to CloudWatch.

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13. What is Detailed Monitoring?
Answer:
Detailed Monitoring provides metrics at a more frequent interval for supported resources such as EC2. It is useful when we need more frequent monitoring data.
For example:
EC2
↓
Detailed Monitoring
↓

More Frequent Metrics

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14. What is CloudWatch Custom Metric?
Answer:
A custom metric is a metric that we send to CloudWatch ourselves instead of relying only on AWS-provided metrics. It is useful for application-specific information.
For example:
A shopping application can send:
SuccessfulOrders = 500
FailedPayments = 10
to CloudWatch as custom metrics.

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15. What is a CloudWatch Namespace?
Answer:
A namespace is a container that separates metrics from different AWS services or applications. It helps organize and identify metrics.
For example:
AWS/EC2
AWS/RDS
AWS/Lambda
Each namespace can contain related metrics.

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16. What are CloudWatch Dimensions?
Answer:
Dimensions are name-value pairs that help identify and filter a metric. They provide additional information about the resource or source of the metric.
For example:
InstanceId = i-123456
This tells us which EC2 instance the metric belongs to.

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17. What is Metric Filter?
Answer:
A Metric Filter can search log data for a specific pattern and convert matching log information into a CloudWatch metric. This is useful for monitoring application errors.
For example:
Application Logs
↓
"ERROR"
↓
Metric Filter
↓
Error Count Metric

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18. What is CloudWatch Logs Insights?
Answer:
CloudWatch Logs Insights is used to search and analyze log data using queries. It helps us find specific errors, requests, or patterns from large amounts of logs.
For example:
We can search application logs to find how many ERROR messages occurred in the last hour.

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19. What is CloudWatch Log Retention?
Answer:
Log retention defines how long CloudWatch Logs should keep log data in a log group. We can choose a retention period based on application and compliance requirements.
For example:
Log Group
↓
Retention = 30 days
↓
Older Logs Removed

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20. What is CloudWatch Composite Alarm?
Answer:
A Composite Alarm combines multiple CloudWatch alarms into one higher-level alarm. This can reduce unnecessary alerts and help create more useful alert conditions.
For example:
CPU Alarm
\
→ Composite Alarm
/
Error Alarm

21. What is CloudWatch Anomaly Detection?
Answer:
Anomaly Detection helps identify unusual metric behavior by learning the normal pattern of a metric. It can detect values that are outside the expected range.
For example:
If an application normally receives 100 requests per minute but suddenly receives 5,000, anomaly detection can identify this unusual behavior.

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22. What is CloudWatch Synthetics?
Answer:
CloudWatch Synthetics allows us to create automated checks called canaries that test applications and endpoints from the outside. It can help verify whether an application is working correctly.
For example:
Canary
↓
Open Website
↓
Check Login
↓
Check Response

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23. What is CloudWatch Application Signals?
Answer:
Application Signals helps monitor application performance and health from an application point of view. It helps us understand whether applications and services are meeting expected performance levels.
For example:
We can monitor application response time and error rate to understand whether users are getting a good experience.

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24. What is CloudWatch Contributor Insights?
Answer:
Contributor Insights helps identify the top contributors to a metric or log pattern. It can help us understand which resources, users, IP addresses, or other dimensions are causing the most activity.
For example:
We can identify which IP addresses are generating the highest number of requests.

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25. What is CloudWatch Logs subscription?
Answer:
A Logs subscription allows log data to be continuously sent to another supported destination for processing or analysis. This is useful when logs need to be processed outside the normal CloudWatch Logs view.
For example:
CloudWatch Logs
↓
Subscription
↓
Lambda / Kinesis / Other Target

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26. How do you monitor EC2 using CloudWatch?
Answer:
CloudWatch automatically receives many EC2 metrics such as CPU utilization and network activity. For additional operating-system metrics such as memory usage, we can install the CloudWatch Agent.
For example:
EC2
↓
CloudWatch
├— CPU
├— Network
└— Memory → Agent

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27. How do you monitor RDS using CloudWatch?
Answer:
RDS automatically sends many database-related metrics to CloudWatch. We can monitor CPU utilization, connections, storage, read/write activity, and other supported metrics.
For example:
RDS
↓
CloudWatch
├— CPU
├— Connections
├— Storage
├— Read

└── Write

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28. What is CloudWatch and SNS connection?
Answer:
CloudWatch can trigger an alarm when a metric crosses a configured condition. The alarm can send a notification through Amazon SNS.
For example:
EC2 CPU > 80%
↓
CloudWatch Alarm
↓
SNS
↓
Email / Notification

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29. What is the difference between CloudWatch and CloudTrail?
Answer:
CloudWatch is mainly used for monitoring metrics, logs, application performance, and alarms. CloudTrail records AWS API activity and helps us understand who performed an action, what action was performed, and when.
For example:
CloudWatch
→ "CPU is high"

CloudTrail
→ "Who deleted this resource?"
Simple ga gurthupettuko:
CloudWatch = What is happening?
CloudTrail = Who did what?

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30. CloudWatch Console — Easy Memory Map
Answer:
Remember CloudWatch like this:

```
CloudWatch
│
├── Dashboards
│   └── Visual Monitoring
│
├── Metrics
│   ├── AWS Metrics
│   └── Custom Metrics
│
├── Alarms
│   ├── Metric Alarm
│   └── Composite Alarm
│
├── Logs
│   ├── Log Groups
│   ├── Log Streams
│   ├── Logs Insights
│   ├── Metric Filters
│   └── Retention
│
├── Events
│   └── EventBridge
│
├── Monitoring
│   ├── Agent
│   ├── Anomaly Detection
│   └── Application Signals
│
├── Synthetics
│   └── Canaries
│
└── Insights
    └── Contributor Insights
```

Simple ga gurthupettuko:
CloudWatch = Metrics + Logs + Alarms + Dashboards + Monitoring + Automation